Telco Customer Churn Analysis Project

TELECOMMUNICATIONS

TEAM NAME: CARAMBOLA

Team Members

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Dataset and Scope

There are 7043 customer records and 21 attributes taken from IBM Watson Analytics

Key Features Analyzed

Demographics: gender, senior citizen, partner, dependents
Services: phone, internet, add-ons
Account: contract, billing, payment method
Financials: monthly charges, total charges, tenure

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CONTEXT & PROBLEM

Business Problem

What customer demographics, service subscriptions, and account attributes drive customer churn in a telecommunications company, and what targeted retention strategies can reduce the churn rate while maximizing customer lifetime value?

Sector Context : Telecommunications .

Stakeholder: Business and retention teams

Objective: Identify churn patterns and improve retention

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TELECOMMUNICATIONS ANALYTICS

Data Engineering

Preparing the IBM Watson Analytics Telco Customer Churn dataset for reliable analysis and retention modeling.

Source

IBM Watson Analytics - Telco Customer Churn dataset

Size and coverage

7,043 customer records × 21 attributes covering demographics, services, account, and financials

Major cleaning steps

- Handle missing values
- Standardize formats
- Validate data types
- Remove duplicates
- Engineer new features

Data dictionary summary

21 attributes grouped into Demographics, Services, Account, and Financials for a structured overview of customer behavior and churn drivers.

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Measuring What Matters

Five KPIs define the churn analysis framework, linking customer behaviour, service usage, and retention impact.

Churn Rate

% of customers who discontinued service.

Revenue at Risk

Monthly revenue exposed to churn from high-value accounts.

Retention Rate

% of at-risk customers retained after intervention.

Customer Lifetime Value (CLV)

Projected long-term value per subscriber.

Conversion-to-Retention Impact

How segmentation and outreach reduce churn risk.

Five KPIs work together to connect churn risk, value at risk, and retention performance.

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Key EDA Insights

Exploratory Data Analysis revealed six critical findings that show the strongest churn drivers and retention opportunities across customer segments, payment methods, and service adoption patterns.

26.5% churn rate

$139K/month revenue at risk.

Month-to-month contracts

42.7% churn vs. 2.8% for two-year contracts.

Electronic check payments

45.3% churn vs. 15-17% for auto-pay users.

5-6 add-ons protect retention

Churn stays below 15% vs. 40%+ with 0 add-ons.

First 12 months are the danger zone

New customers show 47.4% churn.

4 churn segments identified

K-Means clustering shows churn rates from 3% to 44%.

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Advanced Analysis

Unveiling the analytical backbone of our churn prevention strategy.

Statistical Methods

Correlation analysis, chi-square tests, logistic regression, K-Means clustering, and predictive modeling were used to identify churn drivers.

Forecasting

Our analysis forecasts churn risk for customer segments and predicts which customers are most likely to churn in the next period.

Segmentation & Root Cause Findings

K-Means clustering identified 4 distinct customer segments with varying churn rates (3% to 44%). Root cause findings show contract type, payment method, service bundling, and tenure as primary churn drivers.

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Dashboard Overview

Below are the three views that tells what features does our dashboard have: -

Executive View

High-level churn KPIs, trend summaries, and risk hotspots for leadership review.

Operational View

Customer-level and segment-level patterns to support day-to-day retention actions.

Filters & Drilldowns

Interactive slicing by tenure, contract type, service usage, and account attributes.

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TOP INSIGHTS

Four Findings That Change the Game

Decision Insights: Four high-impact, actionable findings that guide retention strategy and resource allocation.

Contract Type is the Strongest Lever

Month-to-month contracts drive **42.7%** churn vs. **2.8%** for two-year contracts. Upgrading contracts is the highest-impact retention tactic.

Service Bundling Dramatically Reduces Churn

Customers with **5-6 add-ons** churn at **<15%** vs. **40%+** with no add-ons. Bundling is a proven retention multiplier.

Payment Method Signals Engagement

Electronic check users churn at **45.3%** vs. **15-17%** for auto-pay. Incentivizing auto-pay enrollment is a quick win.

New Customer Onboarding is Critical

47.4% churn in the first **12 months**. Early engagement and support are essential to survival.

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RECOMMENDATIONS

Turning Telco Insights Into Retention Action

Define the Churn Problem

Use 7,043 customer records and 21 attributes from IBM Watson Analytics to identify what drives churn.

Build the Data Foundation

Complete sourcing, cleaning, EDA, and ETL pipeline with statistical analysis to prepare trusted inputs for KPIs and modelling.

Model Churn Drivers

Analyze demographic, service, account, and financial features to pinpoint the strongest churn predictors.

Deliver Retention Outputs

Turn findings into Tableau dashboards, modelling outputs, and targeted retention strategies.

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IMPACT ASSESSMENT

Impact

This project translates churn analysis into practical retention priorities for telecom teams.

Expected Outcome

- Reduce churn from 26.5% toward target levels through targeted interventions.
- Protect approximately $139K per month in revenue at risk.
- Improve customer lifetime value by retaining more high-value customers.
- Increase overall retention rates with more focused outreach and onboarding.

Priority and Feasibility

- **High Priority, High Feasibility:** Contract upgrades, auto-pay incentives, onboarding programs
- **High Priority, Medium Feasibility:** Service bundling campaigns, segment-specific offers
- **Medium Priority, High Feasibility:** Payment method optimization

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Limitations & Constraints

Data Limitations

- Dataset is from a single telecom company, so it may not generalized to other markets.
- Historical data only; it does not capture recent market changes/trends or new service offerings.
- No external factors included, such as competitor actions, economic conditions, or regulatory changes.
- Limited to 21 attributes, so other unmeasured factors may influence churn.

Method Constraints

- Correlation does not imply causation, so identified patterns may be correlational.
- K-Means clustering assumes spherical clusters and may miss complex segment shapes.
- Predictive models are trained on historical patterns, so future churn may differ.
- Retention strategy effectiveness depends on implementation quality and market conditions.

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Next Steps & Future Preparation

Future Extensions

- Integrate real-time churn prediction into operational systems for proactive intervention(LLM training for real-time prediction)
- Expand analysis to include customer satisfaction scores and NPS data(Net Promoter Score)
- Develop predictive models for churn probability by customer segment
- Test and measure effectiveness of retention campaigns against control groups
- Incorporate external data (competitor pricing, market trends, economic indicators)

Closing Summary

- This analysis provides a data-driven foundation for reducing churn and protecting revenue
- Contract upgrades, service bundling, and improved onboarding are immediate priorities
- Continued monitoring and refinement of retention strategies will maximize impact
- Success requires cross-functional collaboration between analytics, retention, and operations teams

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